

ULETY, Russian Federation

WMO: 308460

Lat: 51.350N

Lon: 112.467E

Elev: 2438

StdP: 13.45

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-22.4	-33.8	0.9	-26.5	-29.1	1.2	-21.6	20.4	7.9	18.7	7.0	0.1	260	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	86.5	62.8	82.7	61.8	79.2	60.8	66.7	78.7	65.1	76.2	63.5	74.0	5.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	94.5	70.8	61.3	88.8	69.3	59.6	83.5	68.0	32.9	79.1	31.5	76.3	30.2	73.9	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.6	14.7	-31.5	92.2	6.1	4.3	-35.9	95.3	-39.5	97.8	-43.0	100.2	-47.4	103.3		
(5)			-31.8	69.8	6.1	2.2	-36.1	71.4	-39.7	72.7	-43.1	73.9	-47.5	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.2	-2.9	5.1	19.8	36.2	49.5	62.4	66.8	61.9	49.6	33.4	13.9	1.2		
	DBStd	25.77	10.45	10.47	10.94	8.27	7.68	6.63	4.89	5.73	7.18	8.61	11.70	10.91		
	HDD50	7563	1638	1256	935	419	105	3	0	2	94	516	1083	1512		
	HDD65	11760	2103	1676	1400	863	483	122	34	128	463	978	1533	1977		
	CDD50	1449	0	0	0	6	89	376	522	371	82	3	0	0		
	CDD65	170	0	0	0	0	2	44	91	32	1	0	0	0		
	CDH74	2262	0	0	0	4	122	732	983	376	45	0	0	0		
	CDH80	658	0	0	0	0	30	220	314	89	5	0	0	0		
	WSAvg	6.0	6.0	6.2	6.3	7.3	7.1	5.4	4.6	4.6	5.6	6.2	6.7	6.2		
	PrecAvg	13.8	0.1	0.0	0.1	0.5	1.2	2.4	3.5	3.5	1.4	0.5	0.2	0.2		
(15) Precipitation	PrecMax	19.9	0.4	0.2	0.4	1.6	4.0	6.0	6.5	8.0	3.5	1.5	0.8	0.5		
	PrecMin	6.7	0.0	0.0	0.0	0.0	0.1	0.1	0.7	0.9	0.0	0.0	0.0	0.0		
	PrecStd	3.2	0.1	0.1	0.1	0.4	1.0	1.3	1.5	1.6	0.9	0.4	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	35.6	54.0	70.1	84.3	91.2	92.1	87.0	79.9	63.9	45.4	30.0		
		MCWB	22.1	27.3	38.9	47.4	56.5	62.8	65.6	63.7	58.3	47.0	35.1	23.9		
	2%	DB	21.3	30.0	47.4	62.9	76.7	86.0	87.3	82.3	73.1	57.4	39.7	24.6		
		MCWB	17.5	23.8	35.1	44.1	53.1	60.9	64.6	63.5	55.2	43.1	31.5	20.4		
	5%	DB	16.0	25.8	42.1	57.2	71.6	81.5	83.3	78.1	67.9	51.6	35.3	20.9		
		MCWB	12.9	20.7	31.7	41.1	50.6	59.6	63.7	62.1	53.0	39.8	28.9	17.5		
	10%	DB	12.3	21.8	37.0	52.3	66.7	77.9	79.4	74.2	63.2	47.0	30.9	17.4		
		MCWB	9.9	17.6	28.7	38.6	48.3	58.6	62.6	61.0	51.2	37.3	26.0	14.7		
	0.4%	WB	22.7	27.6	39.5	48.2	57.5	66.9	69.8	68.3	60.4	47.7	35.4	24.6		
		MCDB	27.0	35.3	53.4	68.7	79.7	81.8	82.5	79.9	73.3	63.2	44.6	29.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	17.5	23.9	35.2	44.7	54.2	63.9	67.4	65.9	57.5	43.5	31.7	20.4		
		MCDB	21.2	29.7	46.9	61.1	73.3	77.3	80.1	75.7	67.5	55.9	39.2	24.3		
	5%	WB	13.1	20.9	31.9	41.8	51.7	62.0	65.9	64.0	55.3	40.7	29.0	17.6		
		MCDB	15.9	25.8	41.6	56.4	68.3	75.1	77.7	73.4	65.0	50.5	35.1	20.8		
	10%	WB	9.8	17.7	28.8	39.1	49.3	60.5	64.5	62.4	52.6	37.9	26.0	14.6		
		MCDB	12.2	21.6	36.7	51.4	65.0	74.0	75.4	71.9	61.9	46.1	30.8	17.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	20.5	22.2	22.4	24.9	23.9	21.4	20.5	21.4	18.9	16.2	15.3		
		MCDBR	18.5	23.9	27.2	32.6	36.3	33.0	30.0	29.0	31.5	27.5	19.3	15.3		
		MCWBR	15.9	19.0	17.7	18.1	16.9	12.9	11.2	12.4	15.2	15.8	13.9	12.8		
		MCDBR	18.4	23.3	26.3	30.4	33.0	24.9	23.6	23.7	26.6	25.9	18.7	15.3		
	5% WB	MCWBR	15.9	18.7	17.3	17.3	16.0	10.6	10.5	11.4	13.8	15.3	13.8	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.280	0.317	0.387	0.481	0.464	0.427	0.388	0.342	0.302	0.276	0.259		
	taud		2.221	2.196	2.191	2.146	1.914	2.078	2.250	2.345	2.417	2.442	2.312	2.184		
	Ebn at Noon		243	267	278	268	246	253	261	266	268	260	233	217		
	Edh at Noon		22	30	36	43	57	49	41	35	29	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		478	838	1341	1660	1850	1932	1782	1536	1218	804	476	364		
	RadStd		13	21	43	122	117	135	145	110	93	49	25	19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page